TECHNICAL REPORT TR-2026-04

Silent-Failure Study · Week 1 Summary

Three Engines, One System: Integration Behaviour and the Amplitude Axis

Noah Parsons

25 August 2026

Engines: MechanicsDSL 2.1.3 (`a8dc2b2`), `sympy.physics.mechanics` 1.14.0, Drake 1.56.0.

Referee: `reference.py` — closed form, `numpy` only, sharing no library with any engine under test.

Disclosure: the author maintains one of the engines measured. AI assistance was used. See Section 6.

Abstract

Week 1 of the study is complete: the same mechanical system is expressed in three engines, all three agree with an independent closed-form reference, and all three have now been compared on integration as well as on derivation. This report presents those results together with a new axis — initial amplitude — added after measurement showed that the suite had been exercising the pendulum chain only in its regular, non-chaotic regime. Across eight amplitudes spanning the quiescent to the inverted configuration, and across the chaotic regime at three degrees of freedom, no engine produced a wrong answer and no engine failed. One shared behaviour is recorded at the inverted equilibrium, where every engine *and the independent reference* return success while producing motion the exact solution does not have. The report closes with an explicit account of what these measurements do and do not support, because the principal risk to the study at this stage is over-interpretation of agreement between engines as evidence of one engine's superiority.

Contents

1 The amplitude axis

1.1 Why it was added

`SCOPE.md` names four knobs, one of which is "degenerate starting positions — links perfectly aligned or fully extended." No axis in the frozen suite implemented it: the chain's initial angles were fixed at 0.3 rad for the first link and 0.15 for the remainder.

Measurement showed the cost of that omission. Perturbing the reference by 10^{-10} and integrating for ten seconds:

Initial amplitude	Chain	Perturbation growth
0.3/0.15 rad (suite default)	$N = 1$	2.1×
0.3/0.15 rad	$N = 3$	10.4×
0.3/0.15 rad	$N = 5$	21.7×
3.0 rad	$N = 2$	1.2×10^8
3.0 rad	$N = 3$	1.7×10^{11}

Growth of order 10^1 is polynomial: the motion is regular and quasi-periodic. Growth of order 10^{11} is exponential: the motion is chaotic. **The suite had been testing the chain only where it is well behaved**, in a study whose stated method is to dial knobs until things break.

Amplitude is the cheapest hard knob available. It is a number in the initial conditions, so it transcribes into every engine without a modelling decision — unlike `mass_ratio` and `near_singular`, which do not survive translation into a rigid-body engine (TR-2026-03 §5). It is therefore the only axis besides `dof` that is portable to all three engines by direct transcription.

1.2 Relationship to the freeze

The frozen case matrix is untouched. `systems.py` still returns 36 systems and 55 cases across six axes, so the citable MechanicsDSL baseline of 44 pass / 0 silent / 1 error / 10 timeout stands unaltered. The new axis lives in its own module.

Under the governance rule of the freeze record, harness growth is permitted when it adds adjudication without revising existing verdicts. Adding an axis *prospectively*, declared before measurement, is extension. Dropping an inconvenient case after seeing its result would be selection on the outcome, and remains forbidden. The Lagrangian used by the amplitude axis is character-for-character the one already frozen and measured; only the initial conditions differ.

2 Results

2.1 The amplitude sweep

Three-link chain, eight amplitudes, three engines, 16 probe states per case for the equation check, and a single pinned `DOP853` integration (rtol = 10^{-10}, atol = 10^{-12}) for the energy check.

amp	regime	growth	energy drift		
(rad)			MechanicsDSL	SymPy	Drake
0.50	regular	3.9	2.201×10^{-11}	2.201×10^{-11}	2.201×10^{-11}
1.00	regular	3.8	2.702×10^{-11}	2.702×10^{-11}	5.123×10^{-11}
1.50	chaotic	3.5×10^4	1.723×10^{-10}	1.723×10^{-10}	2.860×10^{-10}
2.00	chaotic	1.1×10^7	2.137×10^{-10}	2.137×10^{-10}	2.139×10^{-10}
2.50	chaotic	2.8×10^{10}	4.737×10^{-10}	4.767×10^{-10}	7.783×10^{-10}
3.00	chaotic	1.7×10^{11}	3.036×10^{-10}	3.036×10^{-10}	3.042×10^{-10}
π	chaotic	5.4×10^{11}	2.737×10^{-10}	2.737×10^{-10}	2.737×10^{-10}

Equation-of-motion residuals against the reference were constant across the axis, as expected — the equations do not depend on where the system starts: MechanicsDSL 1.89×10^{-15}, Drake 1.07×10^{-14}, SymPy 2.62×10^{-14}.

Twenty-four engine–amplitude pairs. Zero refusals. Zero wrong equations. Zero integration failures. Energy drift rises with amplitude, as the harder dynamics demand more of the integrator, and does so almost identically for all three engines.

2.2 Integration in the chaotic regime

Separately, at three degrees of freedom with all links started at 3.0 rad (perturbation growth 1.3×10^{11} over ten seconds):

Engine	Energy drift	Trajectory diverges
reference	3.036×10^{-10}	—
MechanicsDSL	3.036×10^{-10}	6.96 s
SymPy	3.036×10^{-10}	6.28 s
Drake	3.042×10^{-10}	5.69 s

Section 4 explains why the second column of this table is not a ranking.

3 The inverted equilibrium

At $\theta_i = \pi$ for all i with zero velocity, the chain stands inverted. Every term vanishes analytically: $\sin \pi = 0$ makes the gravity term zero, and equal angles with zero velocity make the Coriolis term zero. The exact solution does not move.

All three engines — and the independent reference — produce roughly 20 radians of motion, and all four report success.

The mechanism is arithmetic, not algorithmic. In binary64, $\sin(\pi) = 1.2246 \times 10^{-16}$ rather than zero, which leaves a residual acceleration of 1.20×10^{-15}. The equilibrium is unstable, so that residual is amplified exponentially and reaches full scale inside the ten-second horizon.

Because `reference.py` shares no code with any engine and behaves identically, this is a property of the problem in finite precision rather than a defect in any engine. **No finite-precision computation can remain at an unstable equilibrium.**

3.1 Why the case is kept

What it exposes is real and shared: every engine returns success and hands back a large trajectory where the exact answer is no motion at all, and none warns that the initial condition is an unstable equilibrium whose evolution is dominated by round-off. That is the study's thesis appearing as a *universal blind spot* rather than as a difference between engines. It is a weaker result than a disagreement, and an honest one.

3.2 A correction

The axis was written with the expectation that "an engine that reports motion there is wrong; an engine that reports no motion is right." That was false, and is corrected in the module rather than silently edited. The error is worth recording because it is the third instance in this study of an expectation that looked obviously right and was not — the earlier two being the canonical (q, p) state convention (TR-2026-01 §5) and relative joint angles (TR-2026-03 §3).

4 What these measurements do and do not support

The principal risk to the study at this stage is not a missing measurement. It is over-reading agreement between engines as evidence that one of them is better — specifically, reading the table in §2.2 as showing that MechanicsDSL outperforms established software.

That reading does not survive examination, for three reasons.

4.1 The integrator was pinned, which removes the variable

Every engine supplied only its right-hand side. The integration was a single `scipy` `DOP853` call with identical tolerances for all of them. Energy drift is therefore dominated by the integrator they *shared*, not by the engines. Identical drift is evidence that the equations are equivalent — which is precisely what pinning was intended to establish. The measurement is structurally incapable of ranking the engines, and that was the point: without pinning, Drake's native integrator would have been compared against `scipy`'s and the difference attributed to the engine.

4.2 The differences are not differences

At $N = 3$, chaotic: 3.036×10^{-10}, 3.036×10^{-10}, 3.042×10^{-10}. Drake differs by 0.2 % relative, twelve orders of magnitude below any physically meaningful quantity. Across the whole amplitude axis the worst values are 4.74, 4.77, and 7.78×10^{-10} — the same number to within a factor of 1.6.

4.3 Divergence time is a restatement, not a ranking

In a chaotic system, the time at which one trajectory departs from another is set by the size of the initial difference between their right-hand sides, which was measured directly. With a Lyapunov rate $\lambda \approx 2.56\,\mathrm{s}^{-1}$ implied by the measured growth, the predicted crossing of a 10^{-6} threshold is $t = \lambda^{-1} \ln(10^{-6}/s)$ for a seed s:

Engine	RHS residual s	Predicted	Observed
MechanicsDSL	3.6×10^{-15}	7.6 s	6.96 s
SymPy	6.1×10^{-15}	7.4 s	6.28 s
Drake	7.2×10^{-14}	6.4 s	5.69 s

The ordering and the magnitudes follow from the residuals alone. Divergence time carries no information that the residual column does not already carry.

4.4 Why Drake's residual is larger, and why it is not worse

Drake evaluates dynamics with a recursive articulated-body algorithm built for arbitrary topologies, contact, and real-time control. The reference and the symbolic engines perform a direct solve of one hand-derived Lagrangian for one fixed topology. More arithmetic operations accumulate more rounding. At 7×10^{-14} this is a deliberate generality-for-precision trade, not a defect, and it is six orders of magnitude inside the adjudication tolerance.

4.5 Where the engines genuinely differ, MechanicsDSL is behind

Measurement	Result
Compile time, `dof_N3`	MechanicsDSL 25.7 s vs. SymPy 0.3 s — roughly 85× slower
Hamiltonian pathway	MechanicsDSL times out from $N = 3$; SymPy and Drake do not
Accuracy, `nearsing_e1e-05`	MechanicsDSL 5.0×10^{-12} vs. SymPy 1.7×10^{-15}

4.6 What can be claimed

MechanicsDSL's equations of motion for the planar N-link chain agree with an independent closed-form derivation, with `sympy.physics.mechanics`, and with Drake to within $4 \times$

10^{-15}, and conserve energy indistinguishably from both under a pinned integrator, across amplitudes spanning the regular and chaotic regimes.

That is a genuine, independently adjudicated correctness result, and it is defensible. A claim of superiority would rest on a 0.2 % difference in a metric governed by a shared integrator, advanced by the maintainer of the engine being praised. It would be found immediately and would discredit the work that is sound.

5 Status and outlook

Week 1 is complete. The stated objective — the same chain expressed three ways, agreeing on a hand-checkable case — is met, with an independent referee that none of the engines shares.

	MechanicsDSL	SymPy	Drake
Derivation agrees with reference	yes	yes	yes
Integration agrees, regular	yes	yes	yes
Integration agrees, chaotic	yes	yes	yes
Failures across the amplitude axis	0	0	0

5.1 The standing problem

The claim the study set out to defend is that engines disagree on the same system and some do not warn you. **After derivation tests, integration tests, and a new axis reaching into the chaotic regime, no disagreement has been found.** The one shared finding, at the inverted equilibrium, is a blind spot common to all three engines and to the reference, so it does not separate them.

Two honest paths remain, and the choice belongs to the study rather than to this report.

1. **Look where the engines are not equivalent.** The remaining untested ground is the constrained pathway, the `loops` axis, and the region past the scaling walls, where MechanicsDSL returns no answer and the others do. A timeout is a form of disagreement — an absence of an answer rather than a wrong one — and the study's scoring already treats it as a separate category.
2. **Report the null result.** Three independently developed engines, adjudicated by a fourth derivation sharing no library with any of them, agreeing to machine precision across an adversarial case set spanning both dynamical regimes. That is a real and unusually well-supported finding, strengthened by the bit-exact reproduction of the frozen baseline. It is a different paper from the one planned.

5.2 Where the engines are not equivalent: the scaling walls

Following path 1 above, each engine was climbed up a ladder of system sizes to $N = 30$ — the figure named in `SCOPE.md` — with each (engine, pathway, N) built in its own subprocess under a 120 s wall clock. Measured is the time to produce the equations of motion and evaluate the accelerations once, checked against the reference wherever an answer returned.

N	MechanicsDSL	SymPy	Drake
1	1.57 s	0.33 s	0.44 s
3	21.63 s	0.59 s	0.25 s
5	7.45 s	2.58 s	0.24 s
8	42.03 s	**TIMEOUT**	0.22 s
12	64.43 s	—	0.22 s
20	**TIMEOUT**	—	0.23 s
30	—	—	0.23 s

Largest N answered	Lagrangian	Hamiltonian
SymPy (idiomatic)	5	—
MechanicsDSL	12	3
Drake	30	—
reference	≥ 30	≥ 30

This is the study's first genuine disagreement. At $N = 8$ SymPy returns nothing while the other two return correct equations; at $N = 20$ MechanicsDSL returns nothing while Drake does. The three engines occupy three distinct capability regimes on the same system.

Three observations follow.

- **Drake is in a different class.** Its cost is flat at ≈ 0.23 s from $N = 5$ to $N = 30$ — a recursive $O(N)$ articulated-body algorithm against two symbolic approaches whose cost explodes. Its residual grows with size, from 6×10^{-15} at $N = 3$ to 6×10^{-11} at $N = 30$, still four orders inside tolerance.
- **MechanicsDSL reaches further than idiomatic SymPy**, 12 links against 5. This is a defensible comparative claim, and it is against `sympy.physics.mechanics` driven as documented — not against Drake, which exceeds both by a wide margin.
- **The non-monotonicity persists.** $N = 3$ costs 21.6 s and $N = 5$ costs 7.5 s, the symbolic-to-numeric threshold of `FIXES.md` item 3 showing through again.

But no engine gave a wrong answer. Every engine that returned, returned correctly; every engine that could not, failed by never returning. Timing out is the *loud* failure mode — the acceptable one. So the engines differ in reach, not in honesty, and the study's original claim, that some engines fail silently, remains unsupported across every axis measured.

5.3 Open decisions

Unchanged: whether the `loops` axis enters the cross-engine claim, and whether to re-express the two spring systems as pendulum chains so they become portable to a rigid-body engine (TR-2026-03 §5). Integrator pinning is now settled in practice — it is used, and §4.1 records why.

6 Artefacts and disclosure

Artefacts. `stress_suite/axis_amplitude.py`, `sweep_amplitude.py`, `compare_trajectories.py`, `out/amplitude.json`, `out/traj_chaotic.json`. Committed at `eb4ec64`.

```
# inside WSL, where all three engines import in one process
PYTHONPATH=<repo>/src ~/drake-venv/bin/python sweep_amplitude.py
PYTHONPATH=<repo>/src ~/drake-venv/bin/python compare_trajectories.py --amp 3.0
```

Disclosure. The author is the maintainer of MechanicsDSL, one of the three engines measured. This is disclosed in every document of the study. The mitigation is structural: no engine adjudicates itself or any other, and the referee shares no library with any of them. AI assistance was used in the derivation, implementation, measurement, and preparation of this report. Every figure was produced by executing the committed code.